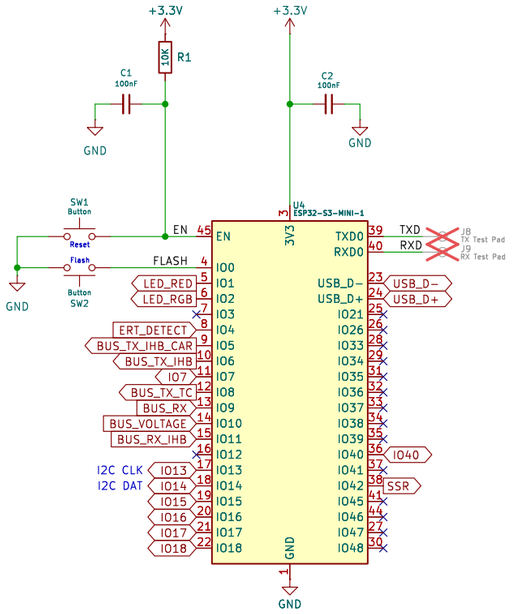
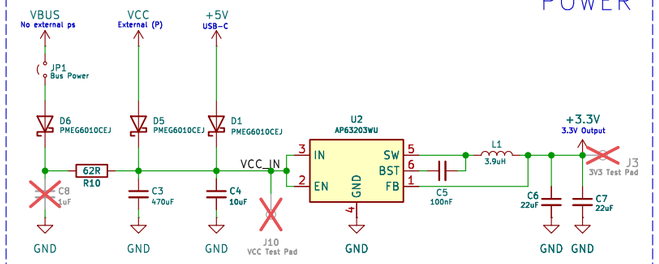
ESP32-S3

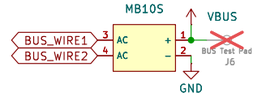


POWER

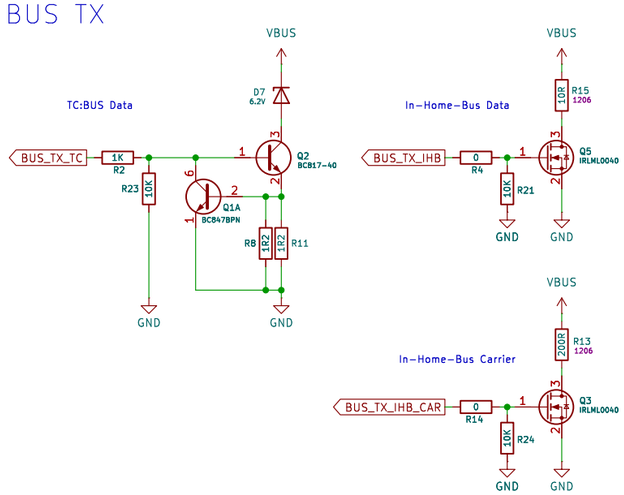


BRIDGE RECTIFIER

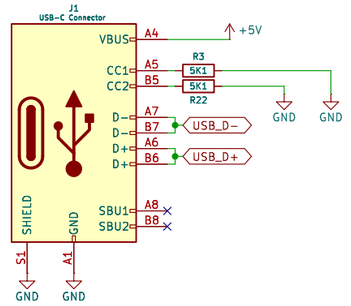
Fix Bus Wire polarity

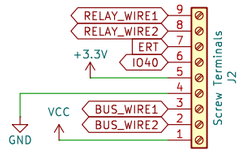


BUS TX

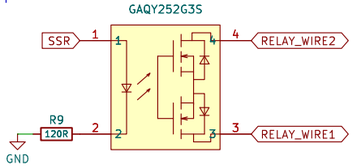


CONNECTORS





RELAY



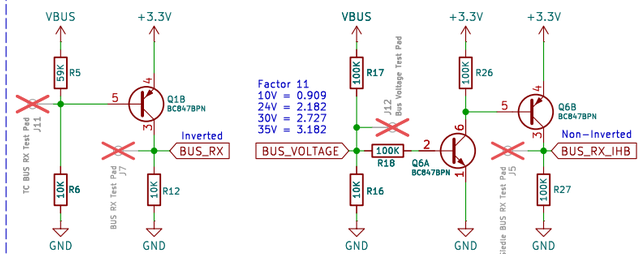
STATUS LEDS



POWER FLAGS



BUS RX / SIGNAL DETECTION



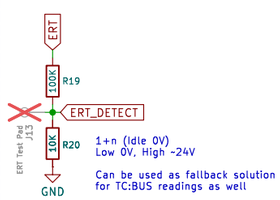
TC:BUS Data(Idle ~22-24V)
Low 8V, High 24V

In-Home-Bus Carrier (Idle ~27-30V)
Low 3V, High 30V(max)

1+n (Idle ~15V)
Low ~15V, High ~24V

Triggers at ~15.5V

In-Home-Bus Data (Idle 27-30V)
Low 2V, High 7V

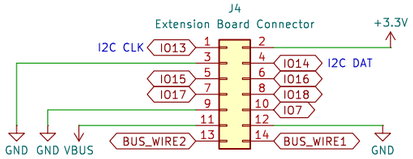


1+n (Idle 0V)
Low 0V, High ~24V

Can be used as fallback solution
for TC:BUS readings as well

J4

Extension Board Connector



Intercom Gateway
For TC:BUS and IN-HOME-BUS based intercom systems

Compatible with TCS (DE), Koch (CH), Scantron (DK), Niko, JUNG (DE) and SIEDLE (DE)

Sheet: /
File: doorman_core_board.kicad_sch

Title: Doorman S3

Size: A3	Date: 2026-05-20	**Rev: 2.2.1**
KiCad E.D.A. 9.0.7		Id: 1/1